

Finite Sample Inference of Individual Treatment Effects

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Abstract

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1 Introduction

Motivation The average treatment effect (ATE) is a common focus in causal inference. This focus on averages is common because of the ease of identifying the parameter, but it could possibly hide heterogeneity in the treatment effects. For instance, if $n = 100$ people enroll in an experiment, 90 of whom experience small and poor effects, but 10 experiences a large positive effect, the ATE would be positive and likely statistically significant.

In this work, we study the analysis of individual treatment effects (ITE). The method we will introduce will allow richer questions to be asked as we will work directly with the distribution of individual treatment effects. By working directly with the collection of treatment effects, we can ask and answer more incisive questions about the effect of treatment. For instance, our method can produce valid confidence intervals for the median treatment effect or any quantile treatment effect. These statistics are more robust to heavy-tailed data and can lead experimenters to recommend better actions.

1.1 Related Work

Caughey et al. [3] motivate much of the work presented here. The authors use randomization inference to test for individual treatment effects being bounded. By inverting these tests, one can derive simultaneous confidence intervals for the individual treatment effects. Their work is insightful, but has some disadvantageous properties. For one, their analysis is valid in the worst-case. Therefore, the confidence intervals can be quite wide and therefore weak. Statistical power is left on the table. Moreover, randomization inference is sensitive to choice of test statistics. Caughey et al. [3] use rank-based statistics for distribution free properties. It is unclear how powerful these test-statistics behave under alternative parameters. The rank statistic is simple, but still has many tuning parameters the analyst can choose. The choice of parameters can lead to unstable results, since some choices are better for different potential outcome distributions.

Fan and Soo Park [8] work in a very similar area as this report, deriving the distribution of treatment effects and its confidence sets. However, the authors assume parametric models

throughout their work, whereas we assume no parametric model for the potential outcomes. Fan and Park [7] give subsequent attention to what they call quantile treatment effects (QTEs), but is now most often called “individual treatment effects” (ITE) as QTEs often denote the difference in marginals, rather than the joint [1, 4, 5]. Fan and Park [7] focus their efforts on deriving asymptotic coverage properties, and invoke a simulation study to understand finite sample properties. The present work differs from Fan and Park [7] in that we guarantee finite sample coverage rates.

We remark here that another subfield of causal inference that attempts to work with heterogeneity is those that focus on conditional average treatment effects (CATEs). For this solution of heterogeneous treatment effects, the analyst has access to attributes of the experimental units. In these subpopulations, analysts can accurately estimate the average treatment effect for this specific subpopulation [2]. This field is useful when the analyst wants to answer what class of units respond well and what class of units don’t. Moreover, the field works closely with machine learning research, to estimate outcome models that are flexible but still able to perform inference on them [17, 10]. Importantly, the CATE is identifiable with observables, whereas the joint distribution of potential outcomes is only partially identified.

Yet the current method has some advantages that the CATE literature does not. At one level, the work we do here is fundamentally different than the CATE. The collection of individual treatment effects is a function of the whole data. We consider heterogeneity with respect to the entire dataset and allow statisticians to answer questions with regard to the whole sample. Another advantage of this method from current the CATE literature is that we do not assume any statistical outcome model, few to little tuning requires, and don’t need bootstrap or asymptotics to quantify uncertainty. In general, this tool is a good addition to the rich research focusing on estimating heterogeneous treatment effects.

2 Set up and Estimand

We consider an experiment on a prespecified n experimental units with two treatment arms. This method can be extended to multiple treatment arms, but for simplicity we work in the binary treatment. Using the potential outcomes framework [9, 12, 15], we define Z_i as the treatment assignment for experimental unit i . With this, $Z_i = 1$ means unit i received treatment whereas $Z_i = 0$ indicates that unit i received control. We write $Y_i(j)$ as the potential outcome that unit i would have had if it had been assigned treatment j . These potential outcomes are fixed but unknown quantities. We can therefore write the observed outcome for experimental unit i as

$$Y_i = Y_i(1)Z_i + Y_i(0)(1 - Z_i).$$

Although the collection $\{Y_i(j)\}_{i=1}^n$ are non-random, there still exists a distribution function for this collections. The true marginal distribution function we write as

$$F_n^{(j)}(t) \triangleq \frac{1}{n} \sum_{i=1}^n 1\{Y_i(j) \leq t\}. \quad (1)$$

This function $F_n^{(j)}(t)$ can be estimated using the empirical distribution of observables. But first, we must make a few assumptions.

Assumption A.1 (No Interference). *No interference means that my treatment assignment does not impact your outcome. In most generality we can write the potential outcome given everyone else's treatment status. This assumption reduces down to*

$$Y_i(Z_1, \dots, Z_n) = Y_i(Z_i).$$

Assumption A.2 (Consistency). *The consistency assumption is also written as “only one treatment version”. It shields against vague treatments that may impact outcomes in non-specific ways. In math we write this as*

$$Y_i = Y_i(1)Z_i + Y_i(0)(1 - Z_i).$$

We note here that assumptions A.2 and A.1 jointly make up what is commonly called the SUTVA, standing for Stable Unit Treatment Value Assumption.

The last assumption we make is equivalent of declaring we are working in a randomized experiment.

Assumption A.3 (Independence of Treatment and Potential Outcomes). *We assume that treatment is independent of the potential outcomes jointly. That is*

$$Z_i \perp (Y_i(0), Y_i(1)), \forall i.$$

This assumption is the crux of causal inference. In the randomized experiment setting, we can safely make this assumption. With this assumption, we can use the empirical CDF of the conditional distribution to estimate $F_n^{(j)}(t)$. O

$$\hat{F}_n^{(1)} = \frac{1}{N_1} \sum_{i=1}^n 1\{Y_i < t \mid Z_i = 1\} = \frac{1}{N_1} \sum_{i=1}^n 1\{Y_i Z_i < t \mid Z_i = 1\} \quad (2)$$

One can check that $\hat{F}_n^{(j)}$ is an unbiased estimator of $F_n^{(j)}$. Moreover, by Glivenko-Cantelli $\hat{F}_n^{(j)} \xrightarrow{a.s.} F_n^{(j)}$.

We switch gears here to discuss individual treatment effects. We define the i th individual treatment effect as

$$\tau_i \triangleq Y_i(1) - Y_i(0),$$

and therefore the distribution of treatment effects is $\{\tau_i\}_{i=1}^n$. We will work with the collection of treatment effects. Our base estimand will be the distribution function of this estimand which we will use $G(t)$ to denote

$$G(t) \triangleq n^{-1} \sum_{i=1}^n 1\{\tau_i \leq t\}.$$

There are two important facts to recognize about $G(t)$. First, $G(t)$ is a fixed but unknown quantity since it is sums of potential outcomes. In addition, the collection of treatment effects is *not* identifiable, meaning we cannot estimate $G(t)$ from the observables. Although $G(t)$ is not identifiable, $G(t)$ can be partially identified. This means that given the observables, $G(t)$ cannot be anything, but instead must lie in some constrained space. If we had access to the marginal distributions exactly $F_n^{(j)}(t)$, we could identify the function $G(t)$ using sharp probabilistic bounds [11, 13]. We say this precisely in the following lemma.

Lemma 2.1 (Makarov Bound). *Assume we only know the marginals $F_n^{(1)}(t), F_n^{(0)}(t)$. Then $G(t)$ must lie in a constrained set that is a function of the marginals.*

$$G_n(t) \in \left(\sup_x F_n^{(1)}(x) - F_n^{(0)}(x - t) \vee 0, \inf_x F_n^{(1)}(x) - F_n^{(0)}(x - t) + 1 \wedge 1 \right)$$

While we do not have the exact marginals $F_n^{(1)}(t), F_n^{(0)}(t)$, we do have estimators for them. We will use these estimators to work backwards, and ultimately obtain a confidence band for $G_n(t)$.

2.1 Randomized Experiments

Before we discuss the main results, we must be clear about the randomization we are working in. The two major types of randomized experiments are Bernoulli Randomized Experiments (BRE) and Completely Randomized Experiments (CRE). In a BRE, each experimental unit is assigned treatment with probability p independently. A consequence is that $N_1 \sim \text{Binom}(n, p)$. In a CRE, we fix N_1 and N_0 (and $N_1 + N_0 = n$). Then each unit is assigned treatment with probability N_1/n . A consequence of this design is that lack of independence of units $\Pr(Z_i = 1, Z_j = 1) \neq \Pr(Z_i = 1) \times \Pr(Z_j = 1)$. This lack of independence between units is a major hindrance for invoking the concentration inequalities we will invoke in the following sections. Therefore, we use a BRE.

A BRE is less common in practice, but much easier to analyze. We assume then that we are in a BRE with fixed probability p , so that $Z_i \perp Z_j, \Pr(Z_i = 1) = p, \forall i$. Extending this method to a CRE requires more work, but can be done.

2.2 Concentration Inequalities for Distribution Functions

The major workhorse of this paper is the celebrated Dvoretzky-Kiefer-Wolfowitz (DKW) inequality [6]. The inequality has many proof techniques often invoking VC-dimensions [14, 16]. All proofs require independent data sampling.

Because we are in a BRE $\{Z_i Y_i(1)\}_{i=1}^n$ are independent. Also, note that $Z_i Y_i(1) = Z_i Y_i$. However, the collection is not identically distributed since our potential outcomes can be anything. Most assumptions assume $Z_i Y_i \stackrel{\text{i.i.d.}}{\sim} F$. However, we assume only that $Z_i Y_i \stackrel{\text{ind}}{\sim} F_i$ are independent, not necessarily identically distributed. It turns out that identically distributed is not necessary to prove DKW results. All we need is independence.

Before we state our theorem, we make one small change to adapt to a BRE. In equation (2), we are dividing by N_1 . In a BRE, this N_1 is a random variable. The proof technique we employ does not work well with random division, so we modify the empirical estimator.

Definition 2.2 (Modified Empirical CDF). We define

$$\hat{F}_{n,p}^{(1)}(t) = \frac{1}{n \times p} \sum_{i=1}^n 1\{Z_i Y_i < t\}. \quad (3)$$

to be our deterministic distribution function estimator.

We give the treated estimator, but the control estimator is similarly defined with $1 - p$ as the denominator. With this definitions, we can introduce a theorem:

Theorem 2.3. *DKW-type for Design Causal Inference*

Let $Z_i \stackrel{i.i.d.}{\sim} \text{Bernoulli}(p)$. Let the potential outcomes be bounded by some constant L . And let $\hat{F}_{n,p}^{(j)}$ be defined as in equation (3) and $F_n^{(j)}$ is equation (1). Then,

$$\Pr \left(\forall x : F_n^{(j)}(x) \in \left(\hat{F}_{n,p}^{(j)}(x) - \sqrt{\frac{8 \log \left(\frac{\alpha}{4} \right)}{-pn}}, \hat{F}_{n,p}^{(j)}(x) + \sqrt{\frac{8 \log \left(\frac{\alpha}{4} \right)}{-pn}} \right) \right) \geq 1 - \alpha.$$

We will call the left end point $\hat{L}_{n,p}^{(j)}$ and the right end point $\hat{U}_{n,p}^{(j)}$.

The concentration inequality we present in Theorem 2.3 is uniform across all x . That is the probability of a single miscoverage event is protected at an α level error. It is this uniformity that is important for downstream inference tasks. Moreover, the inequality is valid for independent, but not identically distributed random variables. The proof of this theorem is in the appendix section A.

2.2.1 Improving The Concentration Inequality

The interested reader may note that Theorem 2.3 is not tight in a various of ways. Skipping this section will not impact further reading. We discuss two ways we can improve this bound and therefore improve the subsequent bounds. The first way the concentration inequality is not tight is through the constant. Inverting Theorem 2.3 gives

$$\Pr \left(\sup_{x \in \mathbb{R}} \left| \hat{F}_n^{(j)}(x) - F_n(x) \right| > \varepsilon \right) \leq 4 \exp \left(-\frac{pn\varepsilon^2}{8} \right).$$

The tight bound can be proven using VC-dimension [14, 16]. The constants would change from $(4, 1/8) \rightarrow (2, 2)$.

$$\Pr \left(\sup_{x \in \mathbb{R}} \left| \hat{F}_n^{(j)}(x) - F_n(x) \right| > \varepsilon \right) \leq 2 \exp \left(-2pn\varepsilon^2 \right).$$

We conjecture that this concentration inequality holds for our fix, but cannot prove it with the tools we can use.

The second and perhaps more interesting way to tighten the bound is to use self-normalized concentration inequalities. A useful fact about distribution functions and quantiles is that they follow a Binomial distribution. Specifically,

$$nq_p \sim \text{Binom}(n, p),$$

where q_p is the p th percentile. The DKW inequality does not use this fact. Therefore, we it is too loose around the edges of the CDF. We see this in the experiments section Figure 1.

3 Main Result

In this section, we give our main result deriving a method to give a uniform confidence band over a distribution of individual treatment effects $\{\tau_i\}_{i=1}^n$. We recall the Makarov lemma 2.1 in which we are able to bound the distribution of τ_i from the marginals alone. To do this, we must find bands for the difference in the marginals themselves. We present this lemma

Lemma 3.1 (Bound on Difference in Marginals). *We define $\Delta(x, t) = F_n^{(1)}(x) - F_n^{(0)}(x - t)$ and $\hat{\Delta}(x, t) = \hat{F}_n^{(1)}(x) - \hat{F}_n^{(0)}(x - t)$. The ε -tube around $\hat{\Delta}(x, t)$ we call $\hat{\Delta}_\varepsilon(x, t) := (\hat{\Delta}(x, t) - \varepsilon, \hat{\Delta}(x, t) + \varepsilon)$ has the property that*

$$\Pr\left(\forall x, \Delta(x, t) \in \hat{\Delta}_\varepsilon(x, t)\right) \geq 1 - \alpha,$$

where $\varepsilon = \varepsilon_1 + \varepsilon_0$ choosing $\varepsilon_1, \varepsilon_0$ to satisfy $\alpha/2$ in our DKW inequality.

In words, the probability that the difference in marginals is contained in the ε -tube around the estimates is greater than $1 - \alpha$, where we pick ε as the sum of $\varepsilon_1, \varepsilon_0$ satisfying $\alpha/2$ in Theorem 2.3. Moreover, the coverage is uniform in x . The proof is found in appendix section A.

With this result, we are able to give our main result on the distribution function of our estimand $\{\tau_i\}_{i=1}^n$.

Theorem 3.2. *Let assumptions A.1, A.2, A.3 hold. Let $G_n(t)$ be the distribution function of the individual treatment effects τ_i . Then*

$$\Pr\left(\forall t, G_n(t) \in \left(\sup_x \hat{L}_{n,p}^{(1)}(x) - \hat{U}_{n,p}^{(0)}(x - t) \vee 0, \inf_x \hat{U}_{n,p}^{(1)}(x) - \hat{L}_{n,p}^{(0)}(x - t) + 1 \wedge 1\right)\right) \geq 1 - \alpha.$$

Proof. The smallest the difference $\hat{\Delta}$ can be is the lower bound of $\hat{F}_{n,p}^{(1)}$ and the upper bound of $\hat{F}_{n,p}^{(0)}$. Likewise for the largest difference. Then, we can apply Makarov inequality. $\square$

The theorem states we can *uniformly* control the distribution function $G_n(t) = \frac{1}{n} \sum_{i=1}^n 1\{\tau_i \leq t\}$ by looking at only the marginal distributions. This error control does not assume any type of distribution, nor does it appeal to asymptotics.

4 Synthetic Data and Experiments

In this section, we illustrate the power of this method by doing some experiments on synthetic data. Our data generating process is described in Table 1.

Since we are simulating the potential outcomes, we know the true collection of individual treatment effects $\{\tau_i\}_{i=1}^n$ by design. We plot the true distribution and the confidence bands we generate around the distribution in Figure 1.

Quantity	Distribution / Definition
Population size	m (fixed)
Treatment assignment	$Z_i \sim \text{Bernoulli}(p)$
Control potential outcome	$Y_i(0) \sim \mathcal{N}(0, 3^2)$
Treated potential outcome	$Y_i(1) \sim \mathcal{N}(3, 3^2)$
Observed outcome	$Y_i = Z_i Y_i(1) + (1 - Z_i) Y_i(0)$

Table 1: Data-generating process for simulations.

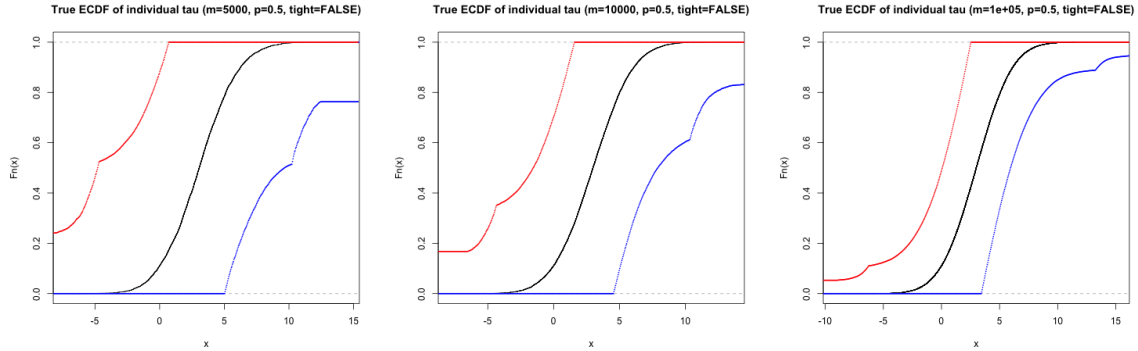


Figure 1: 90% Confidence Intervals for the distribution of individual treatment effects $\{\tau_i\}_{i=1}^n$. As m number of total units in experiment increases, we see tighter bounds.

The confidence bands uniformly cover the true distribution of the individual treatment effects. The kinks one may notice is due to the clipping procedure we employ at the boundaries of the cumulative distribution function. The figures illustrate that we need a lot of samples in order to get reasonable confidence intervals. This weakness is because of our relatively loose concentration inequality stated in Theorem 2.3. Moreover, as $n \rightarrow \infty$, the bounds do not “hug” the true line. This weakness cannot be fixed without stronger assumptions. The weakness is a function that we cannot identify the distribution but can only partially identify it.

We now compare the power of this method with Caughey et al. [3]. We consider constructing confidence intervals for the median treatment effect as a statistic for comparing the two methods.

4.1 Conjecture with Tight Bounds

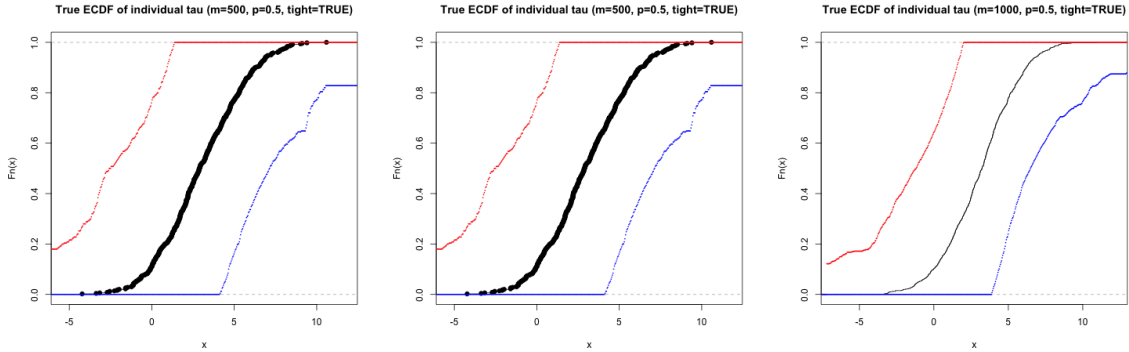


Figure 2: 90% Confidence Intervals for the distribution of individual treatment effects $\{\tau_i\}_{i=1}^n$. Contrast these plots with Figure 1 and we see comparable tightness with much fewer units. This work motivates us to prove this tighter bound, which we believe should hold.

5 Discussion

We provide a closed-form finite sample exact method to give simultaneous bounds on the distribution of individual treatment effects. Previous attempts have focused on randomization inference and asymptotic theory. Our work differs from these previous works in its finite sample exactness and direct calculations. The method is easy to compute and requires no choice of parameters. This new method can help answer questions not typically unanswerable by classical causal inference such as questions on the median treatment effect and any quantile treatment effect.

As we mention through the paper, work on tightening the uniform confidence intervals around marginal distributions will lead to tighter estimates of the individual treatment effect distribution. While the work works in the finite sample setting, we have not shown anything about consistency. Demonstrating consistency and asymptotic behavior could be interesting, though much work in this regime has been done in [7].

I also am interested in applications of this work in adaptive experiments. Waudby-Smith et al. [18] are able to derive “simultaneously valid [confidence sequences] for every quantile of the off-policy reward distribution”. It could be interesting to investigate deriving confidence bands of the counterfactual partially identified difference in reward distributions, rather than marginal off-policy reward distribution. This work is could help start this investigation.

References

- [1] Alberto Abadie, Joshua Angrist, and Guido Imbens. Instrumental variables estimates of the effect of subsidized training on the quantiles of trainee earnings. *Econometrica*, 70 (1):91–117, 2002. Publisher: Wiley Online Library.

- [2] Susan Athey and Guido Imbens. Recursive partitioning for heterogeneous causal effects. *Proceedings of the National Academy of Sciences*, 113(27):7353–7360, 2016. Publisher: National Academy of Sciences.
- [3] Devin Caughey, Allan Dafoe, Xinran Li, and Luke Miratrix. Randomisation inference beyond the sharp null: bounded null hypotheses and quantiles of individual treatment effects. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, 85(5): 1471–1491, 2023. Publisher: Oxford University Press US.
- [4] Victor Chernozhukov and Christian Hansen. An IV model of quantile treatment effects. *Econometrica*, 73(1):245–261, 2005. Publisher: Wiley Online Library.
- [5] Victor Chernozhukov and Christian Hansen. Instrumental quantile regression inference for structural and treatment effect models. *Journal of Econometrics*, 132(2):491–525, 2006. Publisher: Elsevier.
- [6] Aryeh Dvoretzky, Jack Kiefer, and Jacob Wolfowitz. Asymptotic minimax character of the sample distribution function and of the classical multinomial estimator. *The Annals of Mathematical Statistics*, pages 642–669, 1956. Publisher: JSTOR.
- [7] Yanqin Fan and Sang Soo Park. Confidence intervals for the quantile of treatment effects in randomized experiments. *Journal of Econometrics*, 167(2):330–344, 2012. Publisher: Elsevier.
- [8] Yanqin Fan and Sang Soo Park. Partial identification of the distribution of treatment effects and its confidence sets. In *Nonparametric Econometric Methods*, pages 3–70. Emerald Group Publishing Limited, 2009.
- [9] Paul W. Holland. Statistics and Causal Inference. *Journal of the American Statistical Association*, 81(396):945–960, December 1986. ISSN 0162-1459, 1537-274X. doi: 10.1080/01621459.1986.10478354. URL <http://www.tandfonline.com/doi/abs/10.1080/01621459.1986.10478354>.
- [10] Sören R Künzle, Jasjeet S Sekhon, Peter J Bickel, and Bin Yu. Metalearners for estimating heterogeneous treatment effects using machine learning. *Proceedings of the national academy of sciences*, 116(10):4156–4165, 2019. Publisher: National Academy of Sciences.
- [11] GD Makarov. Estimates for the distribution function of a sum of two random variables when the marginal distributions are fixed. *Theory of Probability & its Applications*, 26(4):803–806, 1982. Publisher: SIAM.
- [12] Donald B. Rubin. Estimating causal effects of treatments in randomized and nonrandomized studies. *Journal of Educational Psychology*, 66(5):688–701, 1974. ISSN 1939-2176. doi: 10.1037/h0037350. Place: US Publisher: American Psychological Association.
- [13] Ludger Rüschendorf. Random variables with maximum sums. *Advances in Applied Probability*, 14(3):623–632, 1982. Publisher: Cambridge University Press.

- [14] Bodhisattva Sen. *A Gentle Introduction to Empirical Process Theory and Applications*. Columbia University / Stanford University, 2022. URL <https://sites.stat.columbia.edu/bodhi/Talks/Emp-Proc-Lecture-Notes.pdf>.
- [15] Jerzy Splawa-Neyman, D. M. Dabrowska, and T. P. Speed. On the Application of Probability Theory to Agricultural Experiments. Essay on Principles. Section 9. *Statistical Science*, 5(4):465–472, 1990. ISSN 0883-4237. URL <https://www.jstor.org/stable/2245382>. Publisher: Institute of Mathematical Statistics.
- [16] Jacob Steinhardt. *Lecture Notes for STAT 260 (Robust and Nonparametric Statistics)*. University of California, Berkeley, 2021. URL <https://www.stat.berkeley.edu/~jsteinhardt/stat260/notes/lecture-notes.pdf>.
- [17] Stefan Wager and Susan Athey. Estimation and inference of heterogeneous treatment effects using random forests. *Journal of the American Statistical Association*, 113(523):1228–1242, 2018. Publisher: Taylor & Francis.
- [18] Ian Waudby-Smith, Lili Wu, Aaditya Ramdas, Nikos Karampatziakis, and Paul Mineiro. Anytime-valid off-policy inference for contextual bandits. *ACM/IMS Journal of Data Science*, 1(3):1–42, 2024. Publisher: ACM New York, NY.

A Proofs

Lemma A.1 (Symmetrization Lemma). *Let $\varphi : \mathbb{R} \rightarrow \mathbb{R}_+ \cup \infty$ be convex. If $X_1, \dots, X_n$ are independent on $\mathcal{X}$, $\zeta_i \stackrel{i.i.d.}{\sim}$ Rademacher random variables, and F is a countable family of measurable function from $\mathcal{X} \rightarrow \mathbb{R}$ with $\mathbb{E}[|f(X_1)|] < \infty$. Then*

$$\mathbb{E} \left[\sup_{f \in F} \varphi \left(\sum_{i=1}^n f(X_i) - \mathbb{E}[f(X_i)] \right) \right] \leq \mathbb{E} \left[\sup_{f \in F} \varphi \left(2 \sum_{i=1}^n f(X_i) \zeta_i \right) \right] \quad (4)$$

The proof can be found in most major empirical process textbooks [14].

Proof of Theorem 2.3. By Markov’s inequality $\forall \theta > 0$,

$$\Pr \left(\sup_{x \in \mathbb{R}} \left| \hat{F}_n(x) - F_n(x) \right| > \varepsilon \right) \leq \exp(-\theta \varepsilon) \mathbb{E} \left[\exp \left(\theta \sup_{x \in \mathbb{R}} \left| \hat{F}_n(x) - F_n(x) \right| \right) \right]. \quad (5)$$

$$\sup_{x \in \mathbb{R}_{<0}} \left| \hat{F}_n(x) - F_n(x) \right| = \sup_{f \in F} \left| \sum_{i=1}^n f(X_i) - \mathbb{E}[f(X_i)] \right|.$$

Let our convex function be $\varphi(t) = \exp(\theta t)$ for $\theta > 0$.

$$\begin{aligned} \mathbb{E} \left[\exp \left(\theta \sup_{x \in \mathbb{R}} \left| \hat{F}_n(x) - F(x) \right| \right) \right] &= \mathbb{E} \left[\varphi \left(\sup_{f \in F} \left| \sum_{i=1}^n f(X_i) - \mathbb{E}[f(X_i)] \right| \right) \right] \\ &= \mathbb{E} \left[\sup_{f \in F} \varphi \left| \sum_{i=1}^n f(X_i) - \mathbb{E}[f(X_i)] \right| \right], \end{aligned} \quad (6)$$

where equation 6 is because φ is increasing for $\theta > 0$.

By Lemma A.1.

$$\mathbb{E} \left[\sup_{f \in F} \varphi \sum_{i=1}^n f(X_i) - \mathbb{E}[f(X_i)] \right] \leq \mathbb{E} \left[\sup_{f \in F} \varphi \left(2 \sum_{i=1}^n f(X_i) \zeta_i \right) \right],$$

which for our specific φ yields

$$\mathbb{E} \left[\sup_{f \in F} \exp \left(\theta \sum_{i=1}^n f(X_i) - \mathbb{E}[f(X_i)] \right) \right] \leq \mathbb{E} \left[\sup_{f \in F} \exp(\theta 2 \sum_{i=1}^n f(X_i) \zeta_i) \right].$$

We recall f is of the form

$$f(z) = \frac{1}{pn} b 1\{(-\infty, x]\}(z).$$

Thus we translate

$$\begin{aligned} \mathbb{E} \left[\sup_{b \in \pm 1, x \in \mathbb{R}} \exp \left(\frac{\theta 2b}{pn} \sum_{i=1}^n 1\{(-\infty, x]\}(X_i) \zeta_i \right) \right] &= \mathbb{E} \left[\mathbb{E} \left[\sup_{b \in \pm 1, x \in \mathbb{R}} \exp \left(\frac{\theta 2b}{pn} \sum_{i=1}^n \zeta_i 1\{(-\infty, x]\}(X_i) \right) \mid X \right] \right] \\ &\quad (7) \\ &= \mathbb{E} \left[\mathbb{E} \left[\sup_{b \in \pm 1, x \in \mathbb{R}} \exp \left(\frac{\theta 2b}{pn} \sum_{i=1}^n \zeta_i 1\{(-\infty, x]\}(x_i) \right) \mid X = x \right] \right] \end{aligned}$$

We sort the x so that $x_1 \leq \dots \leq x_n$ and note that

$$\sum_{i=1}^n \zeta_i 1\{(-\infty, x]\}(x_i) = \begin{cases} 0 & x_i > x \\ S_i & x_i < x < x_{i+1} \\ S_n & x \geq x_n \end{cases}$$

where S_i are the running sums of the rademacher random variables.

We recall our first lemma and write

$$\sup_{b \in \pm 1, x \in \mathbb{R}} \exp \left(\theta b 2pn^{-1} \sum_{i=1}^n \zeta_i 1\{(-\infty, x]\}(x_i) \right) = \sup_{b \in \pm 1, 0 \leq i \leq n} \exp(\theta b 2(pn)^{-1} S_i).$$

To get rid of the b we consider either $b = 1, b = -1$

$$\sup_{b \in \pm 1, 0 \leq i \leq n} \exp(\theta b 2(pn)^{-1} S_i) \leq \sup_{0 \leq i \leq n} \exp(\theta 2(pn)^{-1} S_i) + \exp(-\theta 2(pn)^{-1} S_i).$$

Since S_i are symmetric around 0,

$$\sup_{0 \leq i \leq n} \exp(\theta 2(pn)^{-1} S_i) + \exp(-\theta 2(pn)^{-1} S_i) = 2 \sup_{0 \leq i \leq n} \exp(\theta 2(pn)^{-1} S_i). \quad (8)$$

We recall what we're doing in equation (7) and focus on the inner expectation

$$\begin{aligned}
\mathbb{E} \left[\sup_{b \in \pm 1, x \in \mathbb{R}} \exp(\theta 2b(pn)^{-1} \sum_{i=1}^n \zeta_i 1\{(-\infty, x]\}(x_i) \mid X = x) \right] &\leq 2\mathbb{E} \left[\sup_{0 \leq i \leq n} \exp(\theta 2(pn)^{-1} S_i) \right] \\
&= 2\mathbb{E} \left[\exp \left(2\theta pn^{-1} \sup_{0 \leq i \leq n} S_i \right) \right] \\
&\leq 4 \exp(\theta^2 2(pn)^{-1}) \tag{9}
\end{aligned}$$

where equation (9) comes from our original (0th) lemma.

This is independent of the observed values $X = x$, so

$$\mathbb{E} \left[\exp \left(\theta \sup_{x \in \mathbb{R}} \left| \hat{F}_n(x) - F(x) \right| \right) \right] \leq 4 \exp(\theta^2 2(pn)^{-1}) \tag{10}$$

We recall equation 5 is

$$\Pr \left(\sup_{x \in \mathbb{R}} \left| \hat{F}_n(x) - F_n(x) \right| > \varepsilon \right) \leq \exp(-\theta \varepsilon) \mathbb{E} \left[\sup_{x \in \mathbb{R}} \left| \hat{F}_n(x) - F_n(x) \right| \right],$$

which in turn means

$$\Pr \left(\sup_{x \in \mathbb{R}} \left| \hat{F}_n(x) - F_n(x) \right| > \varepsilon \right) \leq 4 \exp(\theta^2 2n^{-1} - \theta \varepsilon).$$

This holds for all θ so minimizing the RHS wrt θ by taking a derivative yields $\theta = \frac{pn\varepsilon}{4}$.
Yielding the final result. $\square$

Proof of Lemma 3.1.

$$\begin{aligned}
&\Pr \left(F_n^{(1)}(x) - F_n^{(0)}(x) \notin \left(\hat{F}_n^{(1)}(x) - \hat{F}_n^{(0)}(x - t) - \varepsilon, \hat{F}_n^{(1)}(x) - \hat{F}_n^{(0)}(x - t) + \varepsilon \right) \right) \\
&\stackrel{\text{Union}}{\leq} \Pr \left(F_n^{(1)}(x) \notin \left(\hat{F}_n^{(1)}(x) - \varepsilon, \hat{F}_n^{(1)}(x) + \varepsilon \right) \right) \\
&+ \Pr \left(F_n^{(0)}(x) \notin \left(\hat{F}_n^{(0)}(x - t) - \varepsilon, \hat{F}_n^{(0)}(x - t) + \varepsilon \right) \right) \leq \alpha
\end{aligned}$$

$\square$